(Please write

END TERM EXAMINATION

(CBCS)(SUBJECTIVE TYPE)

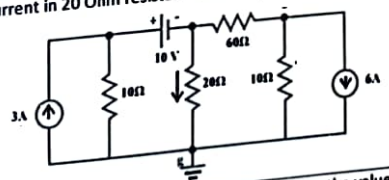
B. Tech (IT/MAE/DMAM), 2nd Semester

(AUGUST, 2022)

Subject Code: BEC 110	**Subject: BASIC ELECTRICAL ENGINEERING**
Time :3 Hours	**Maximum Marks :60**

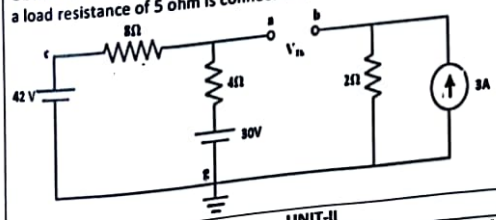
Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

			(2.5*8=20)
Q1	**(a)**	Three resistances 5 Ohm, 8 Ohm and 4 Ohm are connected in parallel to a battery. If current in 4 ohm is 2A then find the total current supplied by the battery?	4-6 A
	(b)	A load of 4 ohms absorbs Maximum power when it is connected to output terminals of a circuit, which has open circuit voltage of 15V at output terminal. Draw the Norton Equivalent of the circuit.	IN = 3.75
	(c)	In a parallel RLC circuit, the current drawn by the resistor, inductor and capacitor are 6A, 12A and 4A respectively. Determine the total current and power factor of the circuit.	I = 10 A, cos φ = 0.6
	(d)	A coil having resistance of 7 ohms and an inductance of 31.8mH is connected to 230V, 50 H supply. Calculate voltage drop across resistor and inductor.	VR = 132.02, VL = 188.31
	(e)	State Flemings Right Hand Rule.	
	(f)	A 400 V line voltage is applied to a balanced 3 phase star load having impedance of 3+j4 ohms per phase. Determine the line current?	
	(g)	A 5kVA, 220/110, 50Hz transformer's primary side (hv side) is connected to 110 V dc. Determine the voltage at secondary of the transformer?	
	(h)	How dc machines are classified on the basis of excitation? Give classification.	
		UNIT-I	
Q2	Find the current in 20 Ohm resistance using superposition theorem?		**(10)**



0.3482 A

Q3	Obtain the Thevenin's equivalent circuit and determine the value of current if a load resistance of 5 ohm is connected at load terminals ab?		**(10)**



		UNIT-II	
Q4	A voltage $V=V_m\sin(\omega t)$ is applied to a load, which draws a current $I=I_m \sin(\omega t+\varphi)$. Derive the expression of average power consumed by the load over a full cycle.		**(10)**

CamScanner

Q5	A coil with a resistance of 10 Ω and an inductive resistance of 5 mH is connected in parallel with another coil having resistance of 10 Ω and inductance of 2mH. The circuit is supplied with a 230 V, 50 Hz ac supply. Calculate i) inductive reactance of both coil ii) overall impedance iii) current drawn from the supply iv) power factor and v) power consumed		**(10)**
		UNIT-III	
Q6	With a neat diagram explain in detail the construction of PMMC instrument. What are shunts and multiplier? Derive the expression for both, with reference to meters used in electrical circuits.		**(10)**
Q7	Explain two wattmeter method for power measurement in a three phase star connected R-L load. Derive the expression of power factor in terms of wattmeter readings. What will be the readings of wattmeter if the load power factor is 0.5?		**(10)**
		UNIT-IV	
Q8	A single phase transformer working at 0.8 power factor has an efficiency of 94% at both 0.75 load and at full load of 600kW. Determine the efficiency at half load and unity power factor?		**(10)**
Q9	Draw a neat diagram of construction of a DC machine. Explain the significance of each part.		**(10)**